Bayesian Inference
Practice Problems

1. Consider a simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, \ldots, n,$$

where Y_i is the response variable, x_i is a fixed covariate, and ϵ_i is the error term corresponding to the ith individual. Assume that $\epsilon_i \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)$, for all i. Here, $\boldsymbol{\beta} = (\beta_0, \beta_1)^\top \in \mathbb{R}^2$ and $\sigma^2 > 0$ are unknown parameters.

Assume prior distributions on $\boldsymbol{\beta}$ and σ^2 as

$$\pi(\boldsymbol{\beta}) \equiv \mathcal{N}_2(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta) \quad \text{and} \quad \pi(\sigma^2) \equiv \texttt{Inv-Gamma}(\alpha_0, \lambda_0),$$

where $\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta, \alpha_0, \lambda_0$ are known hyperparameters. Find the full conditional distributions of $\boldsymbol{\beta}$ and σ^2, and write the Gibbs sampling algorithm.

2. Let X be a single observation and $X|\theta \sim \mathcal{N}(\theta, \sigma^2)$, where σ^2 is known.

   (a) Assume a prior distribution on θ as

$$\pi(\theta) \equiv \texttt{Cauchy}(\mu, \tau),$$

   where μ and τ are known hyperparameters. Write the MH algorithm to generate θ from its posterior distribution.

   (b) Now, assume a prior distribution on θ as

$$\pi(\theta|\lambda) \equiv \mathcal{N}(\mu, \tau^2/\lambda) \quad \text{and} \quad \pi(\lambda) \equiv \texttt{Gamma}(1/2, 1/2),$$

   where μ and τ are known hyperparameters. How do you generate θ using Gibbs sampler? In this context, find the full conditional distributions and write the steps of the sampler clearly.

3. Let a single observation

$$(X_1, X_2) \sim \mathcal{N}_2\left(\begin{pmatrix}\mu_1 \\ \mu_2\end{pmatrix}, \begin{pmatrix}\sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2\end{pmatrix}\right),$$

where $\mu_1, \mu_2 \in \mathbb{R}$ are unknown parameters, and $\sigma_1, \sigma_2 > 0$, $\rho \in (-1, 1)$ are known parameters.

Assume a uniform prior distribution on (μ_1, μ_2) as

$$\pi(\mu_1, \mu_2) \propto 1.$$

Find the full conditional distributions of μ_1 and μ_2, and write the Gibbs sampling steps clearly.